Assignment 1

Personal Development Plan and People Management Skills

Individual written report

First-year master's student

Electronic Information

School of Computer Science

Nanjing University of Information Science and Technology

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Module: [Enter the module title]

Submission date: [Enter the confirmed submission date]

Working draft for student review and personalisation

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Task 1: Personal Development Plan

1. Personal background and development priority

I am a first-year master's student in Electronic Information in the School of Computer Science at Nanjing University of Information Science and Technology. My career objective is to obtain stable employment that makes appropriate use of my technical background. This plan focuses on employability in roles suited to that background. Developing dependable working habits is therefore a practical priority: programming ability is valuable, but employers also need people who can estimate work, meet commitments and coordinate their contribution with others.

I previously participated in a university student innovation project in edge computing, contributing to code editing and the preparation of project proposal documentation. The combination of coding and proposal writing highlights the importance of coordinating activities with different processes and timescales. This experience provides a starting point for considering how I can organise different responsibilities more effectively during postgraduate study.

I regard programming, English and communication as personal strengths, while time management is my main development need. Accordingly, this plan focuses on organising and completing academic and project tasks reliably. The twelve-week schedule starts when I begin implementing it. Baseline performance will be measured at that point; the plan does not assume previous completion rates or missed deadlines.

2. Personal SWOT analysis

Strengths

Programming is a strength that I can develop further through technical assignments and project work. My coding contribution to the edge-computing project provides a concrete example of relevant participation, although I would need additional evidence to demonstrate proficiency in particular languages or tools. I also regard English as a strength, supporting my engagement with technical reading and written explanations. Communication is another advantage when clarifying expectations and coordinating responsibilities. My involvement in proposal writing complements this self-assessment by connecting technical work with the need to present a coherent project rationale. These strengths should help me implement a plan based on clear tasks and documented progress.

Weaknesses

Time management is the weakness I have identified. I need a more consistent method for deciding priorities, estimating effort and protecting time for important work. The exact causes require diagnosis: I should not assume that procrastination, interruptions or unrealistic estimates explain every difficulty. During the baseline period, I will record planned and actual activity to identify recurring patterns. For example, if coding tasks regularly require more time than expected, I can examine whether debugging and testing were included in my estimates. If plans change frequently, I can check whether the original priorities were sufficiently clear.

Opportunities

The first year of a master's programme offers an opportunity to establish effective routines before future responsibilities become more demanding. Current academic tasks can provide realistic practice without creating an unrelated project. My previous exposure to edge computing also gives me a subject area in which to apply planning methods, where suitable coursework or research activities are available. University calendars, library resources and discussions with classmates or tutors may help clarify requirements. A simple calendar, task register and weekly review are sufficient for implementation; paid software or access to specialist hardware is unnecessary.

Threats

Competing deadlines, unexpected technical problems and changes in requirements can disrupt a schedule. An employment search may later introduce additional demands, while excessive attention to immediate requests can displace important longer-term work. A further threat is treating longer working hours as the only solution. That may conceal poor prioritisation and make the routine difficult to sustain. I will manage these risks through contingency time, early clarification of dependencies and realistic workload limits. A late task will be examined for its cause, rather than automatically interpreted as a personal failure.

The SWOT indicates one development priority: reliable time management. My strengths are resources for that improvement, rather than additional weaknesses to address. Better planning should help me use programming, English and communication more consistently, supporting academic progress and preparation for stable employment without suggesting that any personal plan can guarantee a particular job.

3. One SMART goal

My single goal is: **By the end of Week 12, I will achieve an overall on-time completion rate of at least 90% for my registered weekly priority tasks across Weeks 9–12, using a written weekly plan and progress log, with at least five substantive tasks registered each week.**

A priority task will have a defined output, acceptance criterion and planned completion time within that week. Examples include completing and checking a bounded code change, producing an annotated literature summary, or revising a specified proposal section. Tasks should normally represent approximately one to three hours of focused work, allowing meaningful comparison without reducing the register to trivial activities. Larger pieces of work will be divided into deliverables before the weekly plan is fixed. Each completed task must meet its acceptance criterion; an unfinished test or unchecked draft cannot be recorded as complete merely to improve the result.

The completion rate will equal the number of registered tasks completed to the agreed standard by their original planned time, divided by all tasks in that week's fixed register, multiplied by 100. I will combine the counts across Weeks 9–12, rather than average percentages from differently sized lists. At least twenty tasks will therefore be assessed. Changes, cancellations and external delays will remain visible in the original register and be explained separately. I will not remove late tasks retrospectively or reclassify small activities to inflate performance. Formal submission deadlines remain binding regardless of this developmental target.

The goal is specific to dependable task completion, measurable through dated records, relevant to postgraduate work and employment, and time-bound over twelve weeks. Achievability will be reviewed against the baseline and the manageable task size. Doran's (1981) original SMART formulation used assignable and realistic; I use the contemporary achievable and relevant interpretation while retaining personal responsibility. Locke and Latham (2002) show the value of specific goals and feedback, while task complexity and commitment also matter. Consequently, I will interpret results alongside evidence about workload and quality.

4. Action plan, resources and implementation

Weeks 1–2: Establish the baseline

I will record existing academic and project commitments, their deadlines and dependencies. At the beginning of each week, I will register at least five substantive priority tasks, estimating their duration and defining completion. A brief daily record will compare plans with actual activity. At the end of Week 2, I will calculate the baseline completion rate and examine estimation errors. This produces evidence for selecting improvements without inventing a starting score. Resources will be my timetable, current task requirements, a calendar and a spreadsheet or notebook.

Weeks 3–4: Introduce prioritisation and time blocking

I will rank tasks by deadline, importance and dependency, then reserve work periods in my calendar. Research reading or complex coding will receive protected periods rather than being left until smaller requests are finished. I will initially leave approximately 20% of discretionary study time unallocated for delays and new work, adjusting this if records show it is unsuitable. Before accepting an additional commitment, I will identify which existing activity must move. The weekly plan will also separate externally imposed deadlines from personal milestones, so that flexibility does not obscure obligations.

Weeks 5–6: Improve estimates and manage dependencies

I will use earlier records to adjust estimates for similar tasks. For technical work, this means allowing time for checking and debugging; for written work, it includes reading, drafting and revision. If a task depends on feedback or data from another person, I will request it early and prepare a useful alternative activity. In Week 6, I will compare the completion rate, carry-over tasks and estimation errors with baseline results. If the plan remains unrealistic, I will reduce the number of simultaneous commitments or divide large deliverables more clearly.

Weeks 7–8: Stabilise the routine

I will apply the revised method to current responsibilities and practise recognising early warning signs. A task that exceeds its estimate or encounters an unresolved dependency will be reviewed before its deadline. I can use my communication strength to clarify priorities or discuss a revised commitment where appropriate. The original plan and any agreed change will both be retained. This distinguishes responsible adaptation from rewriting the record. I will also review whether the planning process is taking excessive time and simplify unnecessary categories while preserving meaningful evidence.

Weeks 9–12: Evaluate sustained performance

I will continue the same registration and completion rules throughout the final four weeks and calculate the combined on-time rate. The evidence will include weekly registers, dated outputs where appropriate, explanations of delays and a final reflective summary. I will inspect quality alongside completion: for example, whether a code deliverable passed its planned checks or a written section addressed its stated purpose. The target will not justify sacrificing necessary validation. At Week 12, I will compare the final period with the baseline and decide whether the routine is sustainable.

The proposed planning overhead is approximately 100 minutes per week: thirty minutes for weekly planning, ten minutes on each of five study days for logging and adjustment, and twenty minutes for reflection. Academic work itself is scheduled separately according to actual commitments. This is a provisional budget, not a claim about time already available. The routine requires no paid tools, and missed review sessions will be rescheduled promptly.

5. Monitoring, reflection and professional relevance

The progress log will record task description, priority, estimated duration, actual duration, acceptance criterion, planned finish and actual finish. Each weekly review will examine the completion rate and reasons for delays. Estimation accuracy and task quality are supporting diagnostic indicators, not separate development goals. They help explain whether apparent improvement reflects better planning or merely easier tasks.

I will use a planning, monitoring and reflection cycle consistent with self-regulated learning approaches reviewed by Panadero (2017). After a delayed task, I will ask what happened, which factors were controllable and what adjustment is justified. An overlooked testing stage calls for a better estimate; a new external requirement may call for reprioritisation. The response should follow the evidence rather than apply the same explanation to every problem.

Aeon, Faber and Panaccio (2021) associate time management with performance and wellbeing in their meta-analysis, but this does not guarantee that my chosen routine will work. I will therefore examine fatigue and sustainability as well as deadlines. If the target is missed, I will identify the main cause and continue a revised four-week cycle. If it is achieved, I will apply the routine to more demanding postgraduate responsibilities. The intended professional benefit is dependable delivery: using my existing strengths within a realistic, accountable working method that supports my search for stable employment.

Task 2: People Management Skills in Engineering Organisations

1. Introduction

Engineering performance depends on people coordinating specialised knowledge under constraints of cost, time, quality and risk. A technically capable team can still fail when requirements remain ambiguous, responsibilities overlap or employees hesitate to report problems. Effective people management therefore connects individual expertise with collective delivery. This discussion distinguishes leadership from management, examines four management functions, and evaluates communication, leader characteristics, coaching leadership and organisational change. A hypothetical project developing an edge-AI environmental monitoring system provides a consistent example. Such a project brings together electronics, embedded software, machine learning and environmental expertise, making it relevant to Electronic Information studies while illustrating the limits of purely technical solutions.

2. Leadership and management

Management establishes the structures and routines through which an organisation achieves objectives. Leadership influences people to commit to a direction and respond constructively to uncertainty. Kotter (1990) associates management with handling complexity and leadership with handling change. In the monitoring project, management includes assigning tasks, scheduling prototype reviews and checking expenditure. Leadership includes explaining why reliable environmental information matters and helping specialists resolve competing priorities.

These activities overlap rather than belong to separate types of person. A project manager may exercise leadership, while an engineer without formal authority may lead by identifying an overlooked reliability problem. Management without leadership can produce orderly compliance with an unsuitable plan; leadership without management can produce enthusiasm without dependable delivery. Engineering organisations need both. However, inspiring a shared purpose cannot replace technical evidence or justify ignoring safety controls. Influence should support informed judgement rather than pressure employees to accept an unrealistic delivery date.

3. The four functions of management

Planning

Planning translates objectives into requirements, activities, resources and decision points. For the hypothetical system, the team must agree what environmental variables to measure, acceptable prediction performance, power limits and conditions for deployment. Milestones should include requirements review, prototype testing and field evaluation. Planning also identifies dependencies: model development cannot be evaluated meaningfully before data quality and labelling procedures are understood.

A detailed schedule supports coordination, but excessive confidence creates problems when sensor behaviour or model performance remains uncertain. Managers should distinguish confirmed facts from assumptions, provide contingency time and revise plans using test evidence. An apparently efficient schedule that excludes validation can increase rework and undermine trust. Planning should therefore establish measurable outcomes without pretending that every technical difficulty is predictable.

Organising

Organising allocates work, authority and resources. Responsibilities for circuit design, firmware, data processing and validation need clear owners, with explicit agreements about interfaces. A responsibility matrix can identify who makes a decision, who contributes and who must be informed. Documentation and access to shared test equipment also require deliberate coordination.

Specialisation improves depth, but isolated departments can optimise their own outputs at the expense of the complete system. For example, a highly accurate model may consume more power than the embedded platform permits. Cross-functional design reviews help expose such conflicts. Conversely, involving everybody in every decision wastes scarce time. Effective organising combines clear ownership with consultation at points where another discipline's work is affected.

Leading

Leading involves influencing behaviour through purpose, support, feedback and example. The project leader should connect each contributor's work to system reliability and create conditions in which problems are discussed early. When software and hardware teams disagree about a failure, the leader can redirect discussion from blame towards shared evidence and agreed experiments.

Leadership also requires boundaries. Respectful discussion does not mean that all technical opinions have equal evidential strength or that every decision requires consensus. Leaders should explain how decisions are made and allow concerns to be recorded. This supports participation while preserving accountability for timely action.

Controlling

Controlling compares actual performance with agreed standards and initiates corrective action. Suitable indicators include unresolved defects, requirements coverage, power consumption and results on representative data. Schedule and expenditure matter, but completing tasks on time does not establish that the system works safely or reliably. Managers should therefore assess both delivery progress and technical acceptance criteria.

Control becomes counterproductive when employees are rewarded mainly for favourable numbers. Counting closed defects, for example, may encourage superficial fixes or discourage reporting. Managers should investigate trends, inspect evidence and ask what measures overlook. The NIST AI Risk Management Framework emphasises ongoing governance and risk management across the AI lifecycle (NIST, 2023). Control should thus create learning and corrective action, rather than merely produce reassuring dashboards.

4. Communication and active listening

Engineering communication makes assumptions, requirements and decisions understandable across professional boundaries. Terms such as accuracy, failure and real-time performance can mean different things to different specialists. Written interface specifications, concise meeting records and demonstrations reduce ambiguity. A decision record should capture the chosen option, supporting evidence, unresolved assumptions and responsible owner, so that understanding survives staff changes.

Active listening complements documentation. It involves attending carefully, asking clarifying questions, paraphrasing another person's meaning and checking understanding before responding. Kluger and Itzchakov (2022) review how listening can affect workplace relationships and organisational outcomes. In a design review, a leader might ask a firmware engineer to explain when data loss occurs, then restate the problem before proposing a solution. This can reveal that a supposed modelling issue actually originates in sensor timing.

Communication also depends on whether speaking up feels safe. Edmondson (1999) links team psychological safety with learning behaviour. A junior engineer who expects ridicule may conceal uncertainty until correction becomes expensive. Leaders should invite dissent

and acknowledge their own knowledge limits. Nevertheless, psychological safety does not remove performance standards: respectful challenge and responsibility for evidence remain necessary. Teams also need clear escalation routes for urgent safety or security concerns; waiting for the next routine meeting would be inappropriate. Communication quality should be judged by shared understanding and resolved risks, not by the number of meetings held.

5. Characteristics of effective leaders

Effective engineering leaders demonstrate integrity, technical judgement, emotional awareness, fairness and willingness to learn. Integrity means reporting inconvenient test results honestly and applying the same standards to senior and junior staff. Technical judgement enables leaders to ask useful questions and recognise when specialist advice is needed; it does not require being the strongest expert in every discipline.

Self-awareness and emotional regulation help leaders respond proportionately to mistakes. A calm investigation encourages disclosure, whereas public blame can suppress information. Fairness includes distributing demanding work transparently, recognising less visible contributions such as testing, and considering workload before setting deadlines. Goleman (2000) emphasises adapting leadership style to circumstances. No short list of personal traits guarantees success. Leaders must translate values into observable practices, such as explaining decisions, following up commitments and changing direction when evidence contradicts their initial view.

6. Coaching leadership in engineering projects

Coaching leadership develops capability through questions, feedback, supported practice and progressively greater responsibility. Goleman (2000) describes coaching as a style that connects development with organisational needs. In the monitoring project, an experienced engineer might help a junior colleague investigate unstable model performance. Instead of immediately supplying the answer, the coach asks which conditions produce failure, what hypotheses fit the evidence and which experiment would distinguish them. The colleague then proposes and performs a bounded test.

This approach offers three benefits. First, it builds transferable reasoning rather than dependence on one expert. Second, specific feedback makes improvement actionable: an engineer can improve an incomplete test report more readily than respond to a vague judgement about competence. Third, structured development can strengthen motivation

because employees see a credible route to greater responsibility. Coaching is particularly useful where rapidly changing tools make continuous learning essential.

However, coaching requires time, trust and sufficient coach competence. Leading questions can disguise instructions, while poorly informed coaching can reinforce mistakes. Differences in experience also matter: a beginner may need direct explanation before useful reflection is possible. Excessive questioning during an urgent system failure can delay containment and frustrate an otherwise capable engineer. Coaching should therefore be selected for suitable development tasks, with explicit limits on cost, time and risk.

A practical application would begin by agreeing a learning objective, such as independently designing a reproducible test for sensor-data drift. The learner drafts the procedure, the coach reviews assumptions, and both compare the result with acceptance criteria. Responsibility increases only when evidence supports it. Safety-critical releases, security incidents and decisions beyond the learner's competence still require qualified approval. Coaching develops judgement within engineering governance; it should never transfer accountability to someone who lacks authority or preparation.

7. Change management in modern engineering organisations

Engineering organisations face changing requirements, automation, new standards and evolving customer expectations. Introducing an edge-AI platform may alter data ownership, maintenance responsibilities and the skills expected of existing staff. Change management addresses these human and organisational effects alongside the technical implementation. Without it, a technically successful prototype may remain unused or create unreliable working practices.

Kotter (1996) emphasises establishing a reason for change, building support, communicating direction and embedding new practices. Applied here, managers should explain the problem the system addresses, involve engineers and operational users in defining success, and pilot the technology before wider adoption. Training, documented procedures and accessible support help people use the system confidently. Early results should demonstrate useful improvements, while monitoring should also identify additional work or unintended consequences.

Resistance should not automatically be treated as obstruction. Staff may identify valid concerns about reliability, surveillance, workload or loss of expertise. Consultation can improve the change rather than merely persuade employees to accept it. Furthermore, a

staged model cannot predict every dependency in a complex engineering organisation. Managers should combine an overall direction with feedback and reversible trials. Adoption rates alone are insufficient: evaluation should examine system performance, user competence and whether risks remain controlled. Lasting change requires management routines, resources and incentives that support the new practice after initial enthusiasm fades.

8. Conclusion

Effective people management enables engineering expertise to produce coordinated, reliable results. Planning, organising, leading and controlling provide complementary disciplines, while communication and active listening make technical risks visible. Leaders need integrity and adaptable behaviour, and coaching can develop capability when matched to competence and risk. Change management then helps translate technical innovation into sustained organisational practice. Across these activities, the central responsibility is to combine respect for people with evidence, clear accountability and appropriate engineering standards.

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